

TalentFlow

Complete Feature Map v1.0

AI-Assisted Recruitment & Applicant Management Platform

Purpose: Define the complete product feature set before database models and implementation begin.

1. Authentication & Account

Feature	Candidate	Recruiter	Admin	Release
Registration	✓	✓	—	MVP
Login / Logout	✓	✓	✓	MVP
Password reset	✓	✓	✓	MVP
Email verification	✓	✓	—	V2
Role-based access	✓	✓	✓	MVP
Profile management	✓	✓	—	MVP
Account activation/deactivation	—	—	✓	MVP
Session/security management	—	—	✓	V2

2. Candidate Module

Area	Feature	Release	Notes
Profile	Personal information	MVP	Core candidate identity
Profile	Professional summary	MVP	Recruiter-facing profile
Profile	Skills	MVP	Structured skills
Profile	Education	MVP	Education history
Profile	Work experience	MVP	Experience history
Profile	Projects	V2	Portfolio depth
Profile	Certifications	V2	Professional credentials
Profile	Resume upload	MVP	Core document
Profile	Multiple resumes	Advanced	Optional variants
Jobs	Browse / search / filter jobs	MVP	Candidate discovery
Jobs	Job details	MVP	Full job information
Jobs	Save job	MVP	Saved jobs
Applications	Apply for job	MVP	Core workflow
Applications	Application history	MVP	Past applications
Applications	Application status tracking	MVP	Candidate visibility
Applications	Application timeline	V2	Status history UI
Applications	Withdraw application	V2	Candidate control
Experience	Interview schedule	V2	Upcoming interviews
Experience	Offer information	V2	Offer visibility
Experience	Notifications	V2	Recruitment updates

3. Recruiter / Company Module

Area	Feature	Release	Notes
Company	Create company	MVP	Recruiter workspace
Company	Company profile	MVP	Public company identity
Company	Company logo	V2	Branding
Company	Company description	MVP	Company information
Company	Company website	MVP	External company presence
Company	Company locations	V2	Multi-location support
Company	Manage recruiters	V2	Team management
Dashboard	KPI dashboard	MVP	Recruitment overview
Dashboard	Recent applications	MVP	Operational feed
Dashboard	Active jobs	MVP	Job overview
Dashboard	Hiring funnel	V2	Conversion overview
Dashboard	Upcoming interviews	V2	Interview operations
Dashboard	Advanced analytics	Advanced	Deeper recruitment intelligence

4. Job Management

Feature	Release	Notes
Create job	MVP	Core recruiter action
Edit job	MVP	Update open roles
Draft job	MVP	Prepare before publishing
Publish / unpublish	MVP	Control visibility
Close job	MVP	End recruitment
Job status	MVP	Draft, published, closed
Employment type	MVP	Full-time, part-time, etc.
Experience level	MVP	Required seniority
Salary range	MVP	Compensation information
Location	MVP	Physical or geographic location
Remote / Hybrid / On-site	MVP	Work arrangement
Required skills	MVP	Core matching input
Preferred skills	V2	Additional matching input
Job duplication	V2	Faster job creation
Job performance statistics	V2	Applications and conversion

5. Applicant Management

Feature	Release	Notes
Applicant list	MVP	Recruiter view
Search applicants	MVP	Find candidates
Filter applicants	MVP	Narrow candidate pool
Applicant profile	MVP	Candidate information
Resume viewing / download	MVP	Recruiter document access
Application details	MVP	Job-specific application
Candidate skills / experience	MVP	Structured candidate data
Application date / status	MVP	Operational context
Recruiter notes	V2	Internal notes

Feature	Release	Notes
Candidate rating	V2	Evaluation aid
Shortlist candidate	MVP	Pipeline action
Reject candidate	MVP	Pipeline action

6. Recruitment Pipeline

Feature	Release	Notes
Application stages	MVP	Applied → Screening → Shortlisted → Interview → Offer → Hired
Stage transitions	MVP	Controlled movement
Kanban pipeline	MVP	Visual candidate board
Drag & drop	V2	Fast stage movement
Status history	MVP	Track every transition
Stage-specific actions	V2	Rules by stage
Candidate evaluation	V2	Structured assessment
Bulk candidate actions	V2	Recruiter productivity

7. Interview Management

Feature	Candidate	Recruiter	Release
View interviews	✓	✓	V2
Schedule interview	—	✓	V2
Reschedule / cancel	—	✓	V2
Interview round	—	✓	V2
Interview mode	—	✓	V2
Meeting link	✓	✓	V2
Interview reminders	✓	✓	Advanced
Interview feedback	—	✓	V2
Interview score	—	✓	V2

8. Offer Management

Feature	Release	Notes
Create offer	V2	Recruiter creates offer
Salary information	V2	Compensation details
Joining date	V2	Expected start
Offer status	V2	Track offer state
Send offer	Advanced	Delivery workflow
Candidate offer view	V2	Candidate access
Accept / reject offer	Advanced	Candidate response
Offer history	Advanced	Auditability

9. Notification System

Feature	Release	Notes
Notification model	V2	Persistent notification records
In-app notifications	V2	Core notification channel

Feature	Release	Notes
Read / unread status	V2	Notification state
Notification center	V2	Central inbox
Email notifications	Advanced	External delivery
Interview reminders	Advanced	Scheduled reminders
Background notification processing	Advanced	Celery-based processing

10. Resume Management

Feature	Release	Notes
Resume upload	MVP	Store candidate resume
Resume storage	MVP	Secure document storage
Resume download	MVP	Authorized recruiter access
Resume preview	V2	Browser-friendly viewing
Resume parsing	Advanced	Extract structured information
Skill extraction	Advanced	Identify skills
Structured profile extraction	Advanced	Populate candidate profile

11. Intelligent Matching

Feature	Release	Notes
Skill matching	Advanced	Compare candidate skills with job requirements
Match percentage	Advanced	Explainable score
Missing skills	Advanced	Identify skill gaps
Experience matching	Advanced	Compare experience against requirements
Candidate ranking	Advanced	Rank applicants for a role
AI-assisted candidate insights	Advanced	Optional decision-support layer

12. Analytics & Reporting

Feature	Release	Notes
Basic KPIs	MVP	Jobs, applicants, interviews, hires
Application statistics	V2	Application trends
Hiring funnel	V2	Stage conversion
Charts	V2	Visual analytics
Job performance	V2	Per-job metrics
Time-to-hire	Advanced	Hiring efficiency
Advanced reports	Advanced	Deeper analytics
Export reports	Advanced	Portable reporting

13. Admin Panel

Feature	Release	Notes
User management	MVP	Platform administration
Company management	MVP	Platform administration
Job management	MVP	Moderation / oversight
Application monitoring	MVP	Platform visibility

Feature	Release	Notes
User activation / deactivation	V2	Account control
Platform analytics	V2	System-wide metrics
Audit logs	Advanced	Trace sensitive actions
Reported content	Advanced	Moderation workflow

14. Audit & Activity System

Feature	Release	Notes
Activity tracking	V2	Operational history
Status history	MVP	Candidate stage history
Audit log	Advanced	Security and compliance trail
Admin audit view	Advanced	Central audit inspection

15. API Layer

Feature	Release	Notes
REST API	MVP	React ↔ Django communication
Authentication API	MVP	Auth endpoints
Job API	MVP	Job CRUD and discovery
Application API	MVP	Application workflow
Candidate API	MVP	Candidate data
Interview API	V2	Interview operations
Analytics API	V2	Dashboard metrics
API documentation	Advanced	Developer usability
API versioning	Advanced	Long-term API stability

16. Background Processing

Feature	Release	Notes
Email notifications	Advanced	Async delivery
Interview reminders	Advanced	Scheduled tasks
Resume processing	Advanced	Async document handling
Resume parsing	Advanced	CPU/API-heavy processing
Analytics calculations	Advanced	Background aggregation
Matching calculations	Advanced	Async scoring

17. Security & Permissions

Feature	Release	Notes
Authentication	MVP	Identity verification
Role-based permissions	MVP	Candidate / recruiter / admin
Object-level permissions	MVP	Prevent unauthorized object access
Secure file validation	MVP	Protect resume/document uploads
Environment secrets	MVP	Keep credentials outside source
API authentication	MVP	Protect API
CSRF protection	MVP	Request protection

Feature	Release	Notes
Input validation	MVP	Validate user data
Audit logging	Advanced	Trace critical actions
Security testing	Advanced	Verify security controls

18. Testing

Feature	Release	Notes
Model tests	MVP	Core domain behavior
Service / business logic tests	V2	Application rules
API tests	V2	Endpoint behavior
Permission tests	V2	Authorization behavior
Integration tests	Advanced	Cross-module workflows
Regression testing	Advanced	Protect releases

19. Deployment & DevOps

Feature	Release	Notes
Environment variables	MVP	Configuration management
Production settings	V2	Secure deployment config
Docker	Advanced	Reproducible environments
Docker Compose	Advanced	Multi-service local/prod setup
CI/CD	Advanced	Automated delivery
Production deployment	Advanced	Live application
Logging / monitoring	Advanced	Operational visibility

Final Product Architecture

TALENTFLOW

Candidate → Profile · Resume · Jobs · Applications · Interviews · Notifications
Recruiter → Company · Jobs · Applicants · Pipeline · Interviews · Analytics
Admin → Users · Companies · Jobs · Monitoring · Analytics · Audit

All major workflows communicate through a versioned REST API.
React frontend → Django + Django REST Framework → PostgreSQL
Optional asynchronous layer → Redis → Celery workers

Core recruitment workflow:
Create Job → Publish Job → Applications → Screening → Shortlisted → Interview → Evaluation → Offer → Hired

Progressive Development Strategy

Phase	Scope	Priority
Phase 1 — Foundation	Authentication, roles, company, candidate profile, jobs	MVP
Phase 2 — Recruitment Core	Applications, resume, applicant management, pipeline, status history	MVP
Phase 3 — Professional Features	Interviews, feedback, notifications, analytics	V2
Phase 4 — Engineering	Celery, Redis, email, audit logs, advanced testing	Advanced
Phase 5 — Intelligence	Resume parsing, skill extraction, candidate-job matching, AI insights	Advanced
Phase 6 — Production	Docker, CI/CD, deployment, documentation, portfolio	Advanced

Key Engineering Rule

Do not add technology just to make the stack look impressive. Every component must solve a real product problem. Redis, Celery, AI, Docker, CI/CD and advanced analytics are introduced when the product actually benefits from them. The core recruitment system must remain useful even without AI.